

curriculum vitae of
Yaroslav Zharov
LEAD OF ML RESEARCH · AI AGENTS
MUNICH, GERMANY

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I have worked on neural networks since 2013 and published my first papers on them in 2015. Since then my aim has been machine learning that people actually use: in security and banking (Kaspersky, Sberbank), in X-ray imaging for physicists and biologists (KIT, PhD at Heidelberg), and in AI agents for software engineering at JetBrains, where I grew a research group from one report to about 36 people. I try to understand a problem as deeply as it needs, to stop my own ideas when the evidence is against them, and to build teams in which people grow into leads.

WORK EXPERIENCE

2023 – now

Lead of ML Research (2026 – now); Research Team Lead (2023 – 2026) JETBRAINS RESEARCH

- Grew a research team from one report to nine people. Since August 2026, as Lead of ML Research, I lead the team leads: about 36 people, direct and indirect. Several team members now lead projects and research directions of their own.
- Set the team's research line on agents for software engineering: environments (*EnvBench*), training (*PIPer*: with SFT and RLVR, *Qwen3-8B* matches *Qwen3-32B* and *GPT-4o* at environment setup; *SRFT*: learning from unresolved trajectories, +3.7 pp on SWE-bench Verified), context management (*The Complexity Trap*: plain observation masking halves agent cost at the solve rate of LLM summarisation), and evaluation (*GitGoodBench*, software world models).
- Built an agent-platform prototype with the team and used it in several projects with product teams. Took it to production together with two team members; the production platform, rewritten for product needs, is now Koog.
- Research delivered with products: the notebook error-resolution agent in Datalore (EMNLP'24 demo), the create-project assistant with the Rider team, and offline metrics for commit message generation that match online user behaviour, with the AI Assistant evaluation team (ICSE-SEIP'25).
- Stopped a direction on per-step model routing for agents after three to four months and three projects, once a random model-mixing baseline proved too hard to beat; the people moved to other work.
- Ran two company-wide series, on building AI agents and on evaluating ML features, for about 200 colleagues.

2023

Visiting Researcher REXPLORER

Turned a business request into an ML task and built the labelling, training, and inference pipelines for rooftop solar panel segmentation. The pipeline went into use; the project continued after the start-up phase.

2019 – 2023

Researcher, PhD Student KARLSRUHE INSTITUTE OF TECHNOLOGY, IPS

- Self-supervised learning for X-ray tomography: label-free denoising, pre-training, sample localisation, labelling optimisation, and inpainting of lost data. The main outcome for the users was less time spent on manual labelling.
- For shot noise reduction (Optics Express 2023), worked through the imaging process layer by layer to design the training prior. Cleaner images at lower dose mean shorter synchrotron time for the same result.
- Built training and inference pipelines for very large tomographic volumes; supported biologists and physicists with classical CV and deep learning in daily work.
- Organised a labelling workshop to grow the labelled data for the main project.

2017 – 2019

Researcher SBERBANK AI LAB

- Research on interpreting neural networks (YASENN, with A. Tuzhilin); two patents granted.
- Random forests and interpretation methods for credit scoring; time series analysis for security incident detection; statistical consulting for colleagues outside ML.

2016 – 2017

Analyst SBERBANK TECHNOLOGY, ML LAB

Text classification for internal correspondence (deployed), named entity recognition, time series forecasting.

2014 – 2016	Analyst-Developer	KASPERSKY LAB R&D
	Built the parser, website, and internal system behind threats.kaspersky.com , still in service. Built a system for preparing, storing, and analysing user statistics; prototyped detection of potentially unwanted installations.	
2013 – 2015	Trainee	INSTITUTE OF CONTROL SCIENCES, RUSSIAN ACADEMY OF SCIENCES
	Wavelet neural networks and initialisation of multilayer and convolutional networks; two papers in 2015.	

EDUCATION

2021 – 2025	PhD in Computer Science, magna cum laude	HEIDELBERG UNIVERSITY
	Thesis: <i>Relaxing Supervision Requirements for Tomographic Data Analysis with Machine Learning</i> .	
2010 – 2015	Dipl.-Ing. in Applied Mathematics	MOSCOW ENGINEERING PHYSICS INSTITUTE (MEPhI)
	Control systems, digital signal processing, computer vision, programming.	

SELECTED PUBLICATIONS

About 30 papers since 2015; full list on Google Scholar.

- T. Lindenbauer, I. Slinko, L. Felder, E. Bogomolov, **Y. Zharov**. The Complexity Trap: Simple Observation Masking Is as Efficient as LLM Summarization for Agent Context Management. *DL4Code workshop at NeurIPS 2025*.
- A. Kovrigin, A. Eliseeva, K. Grotov, E. Bogomolov, **Y. Zharov**. PIPer: On-Device Environment Setup via Online Reinforcement Learning. *arXiv 2025* (workshop version at DL4Code, NeurIPS 2025).
- I. Slinko, I. Zavidnyi, E. Bogomolov, **Y. Zharov**. Step Rejection Fine-Tuning: A Practical Distillation Recipe. *arXiv 2026*.
- E. Bogomolov, **Y. Zharov**. Towards Evaluation of Implicit Software World Models in Coding LLMs. *DL4Code workshop at ICML 2026*.
- A. Eliseeva, A. Kovrigin, I. Kholkin, E. Bogomolov, **Y. Zharov**. EnvBench: A Benchmark for Automated Environment Setup. *DL4Code workshop at ICLR 2025*.
- P. Tsvetkov, A. Eliseeva, D. Dig, A. Bezzubov, Y. Golubev, T. Bryksin, **Y. Zharov**. Towards Realistic Evaluation of Commit Message Generation by Matching Online and Offline Settings. *ICSE-SEIP 2025*.
- K. Grotov, A. Borzilov, M. Krivobok, T. Bryksin, **Y. Zharov**. Debug Smarter, Not Harder: AI Agents for Error Resolution in Computational Notebooks. *EMNLP 2024, System Demonstrations*.
- **Y. Zharov**, E. Ametova, R. Spiecker, T. Baumbach, G. Burca, V. Heuveline. Shot Noise Reduction in Radiographic and Tomographic Multi-Channel Imaging with Self-Supervised Deep Learning. *Optics Express 31(16)*, 2023.
- **Y. Zharov**, D. Korzhnikov, P. Shvechikov, A. Tuzhilin. YASENN: Explaining Neural Networks via Partitioning Activation Sequences. *arXiv 2018*.
- D. Kataev, **Y. Zharov**. Usage of Wavenets of Arbitrary Dimensionality for Objects Modelling. *IMMOD 2015*.

PATENTS

- Method of debugging a trained recurrent neural network. WO 2020167156 A1, RU 2715024 C1.
- Method of interpreting artificial neural networks. WO 2020013726 A1, RU 2689818 C1.

SERVICE AND TEACHING

- Reviewer: ICML 2026 (Gold Reviewer), NeurIPS 2026, AAAI 2026.
- Invited talk on AI agents, IDE Workshop at ICSE 2025.
- Supervised about seven Bachelor's and Master's theses with DFKI, TUM, LMU, and Constructor University Bremen.

SKILLS

Research: LLM agents for software engineering, post-training (SFT, RLVR, rejection fine-tuning), evaluation and benchmark design, self-supervised learning, computer vision, interpretability, statistics.

Engineering: Python, PyTorch, Hugging Face Transformers, NumPy, Linux, Git, $\LaTeX$.

Languages: Russian (native), English (fluent), German (B2).

References available on request.